Measures of Spread in Statistics

Understanding how data spreads around its centre is crucial for statistical analysis. We'll explore the key measures that reveal data variability and distribution patterns.

What Are Measures of Spread?

Whilst measures of central tendency show us the "middle" of a dataset, measures of spread reveal how our data points are distributed around that centre. Also known as measures of dispersion or scatter, these statistics tell us whether our data is tightly clustered or widely scattered.

The four primary measures of spread are range, interquartile range (IQR), variance, and standard deviation. Each provides unique insights into data variability.

Range: The Simplest Measure

Definition

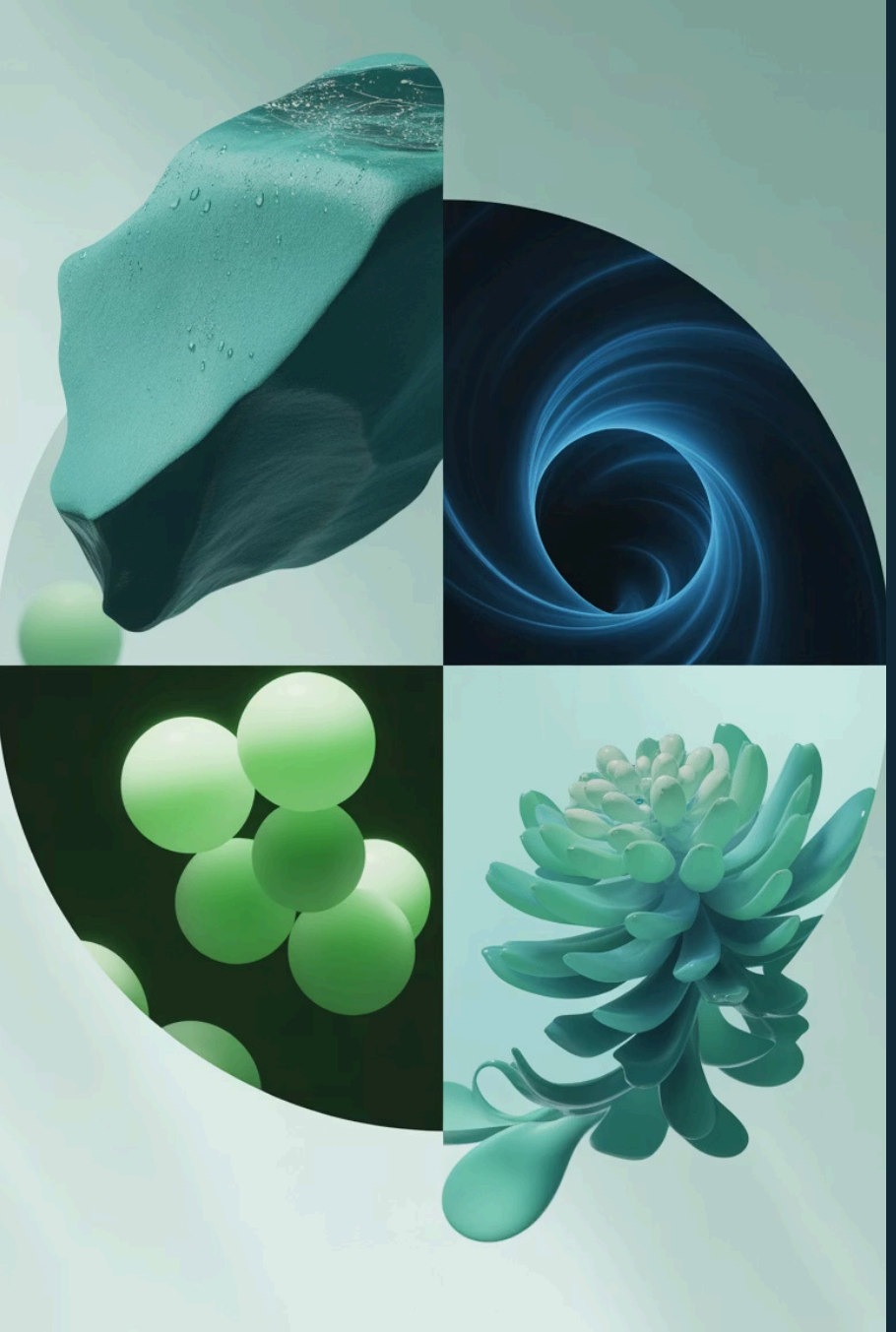
Range = Maximum value - Minimum value

Golf Scores Example

From the stem-and-leaf plot:

- Lowest score: 66
- Highest score: 75
- Range: 75 - 66 = 9

The range provides a quick snapshot of data spread but can be heavily influenced by extreme values or outliers.



Understanding Quartiles

Think of dividing something into quarters - you cut it in half, then cut each half again. Similarly, we can divide datasets into quarters using quartiles.

01

First Quartile (Q_1)

The median of the lower half of the data - separates the lowest 25% from the rest

02

Second Quartile (Q_2)

The median of the entire dataset - divides the data into two equal halves

03

Third Quartile (Q_3)

The median of the upper half of the data - separates the highest 25% from the rest

Calculating Interquartile Range (IQR)

The interquartile range measures the spread of the middle 50% of data points.

IQR = Q_3 - Q_1

Golf Scores Example (18 data points)

Data: 66, 67, 67, 68, 68, 68, 68, 69, 69, 69, 69, 70, 70, 71, 71, 72, 73, 75

- Median (Q_2): 69
- Q_1 (median of lower half): 68
- Q_3 (median of upper half): 71
- IQR: 71 - 68 = 3

IQR with Odd Number of Data Points

Example: 11 Golf Scores

Data: 66, 67, 68, 69, 69, 69, 70, 71, 71, 72, 75

Median: 69

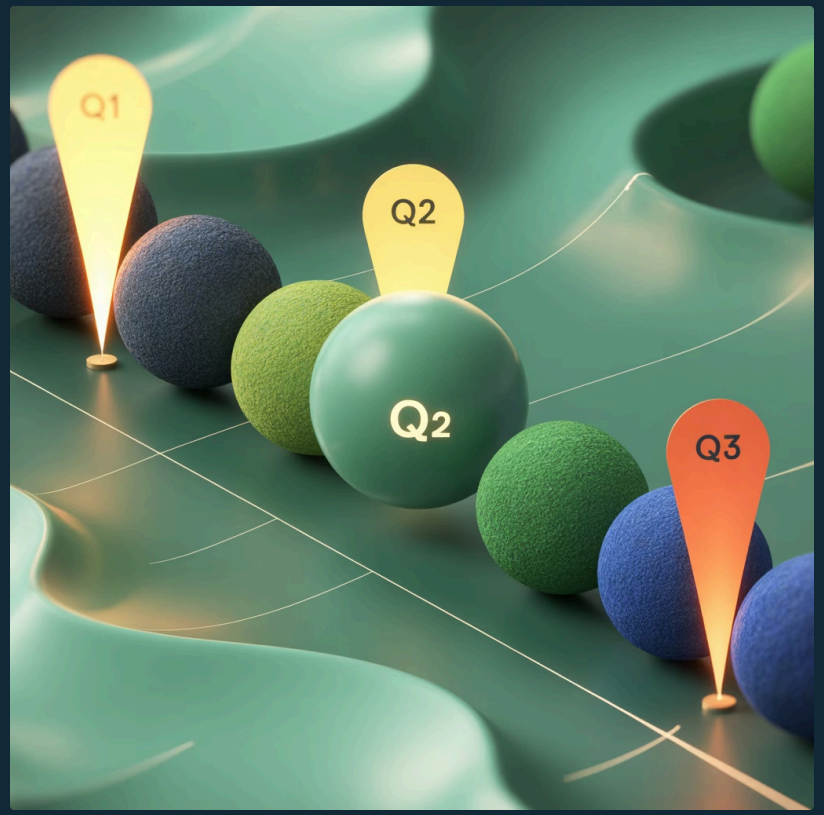
Lower half (excluding median): 66, 67, 68, 69, 69

$Q_1 = 68$

Upper half (excluding median): 70, 71, 71, 72, 75

$Q_3 = 71$

IQR = 71 - 68 = 3



When the dataset has an odd number of points, we exclude the median when calculating Q_1 and Q_3.

How Data Changes Affect Measures

Shifting (Adding/Subtracting Constants)

Adding 6 to each value in {3, 3, 7, 9, 13}:

- Mean: 7 → 13 (+6)
- Median: 7 → 13 (+6)
- Range: 10 → 10 (unchanged)
- IQR: 8 → 8 (unchanged)

Scaling (Multiplying/Dividing)

Multiplying each value by 2:

- Mean: 7 → 14 (×2)
- Median: 7 → 14 (×2)
- Range: 10 → 20 (×2)
- IQR: 8 → 16 (×2)

Adding or Removing Data Points

Effects on Mean vs Median

Original data: {1, 2, 3}

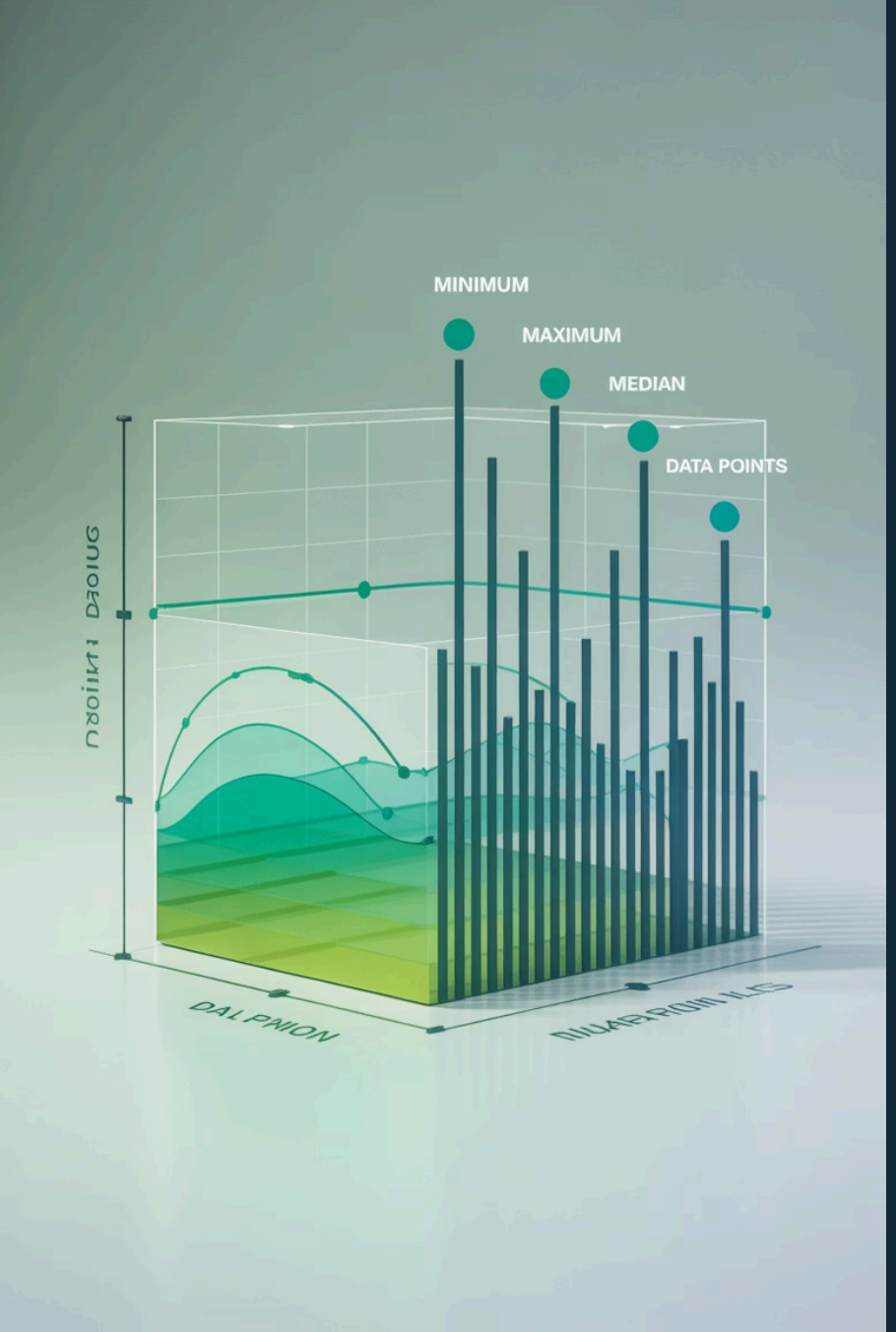
- Mean: 2
- Median: 2

Adding extreme value 1,000:

- Mean: 252 (massive change)
- Median: 2.5 (minimal change)

The median is more resistant to outliers than the mean, making it a better measure of central tendency for skewed data.

Outliers are extreme values that can dramatically affect statistical measures, particularly the mean.



Box-and-Whisker Plots

Box plots provide a visual summary of data distribution, showing the five-number summary at a glance.

The Box

Contains the middle 50% of data. Left edge is Q_1, right edge is Q_3. The line inside shows the median.

The Whiskers

Extend to the minimum and maximum values, showing the full range of the data.

Quick Insights

Instantly see the median, range, IQR, and data distribution symmetry.

Five-Number Summary

Every box plot is based on the five-number summary, which provides a complete overview of data distribution.

Minimum	Q_1	Median	Q_3	Maximum
2	5	11	13	14

From this summary, we can quickly calculate:

- Range: 14 - 2 = 12
- IQR: 13 - 5 = 8
- Data distribution across quartiles

Understanding measures of spread empowers you to make informed decisions about data variability and choose appropriate statistical methods for analysis.